The empirical recurrence for OEIS A252453, recovered from its tail

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Abstract

OEIS A252453 records "Empirical recurrence of order 69" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 109$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 5$, so the recurrence is proved for every $n \geq 74$. Its last coefficient is $c_{69} = 288 \neq 0$, so no further tail satisfies anything shorter; any order-69 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A252453 is "Number of (n+2)X(4+2) 0..3 arrays with every 3X3 subblock row, column, diagonal and antidiagonal sum not equal to 0 2 4 6 or 7.". It has offset 1 and begins

$$747, 2643, 11334, 48170, 203320, 863351, 3675259, 15672575, \ldots$$

The entry states:

> Empirical recurrence of order 69 (see link above)

As of the "Last modified" line on the live entry (Jun 02 2025 11:16:04, revision 8) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 109$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 109$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. □

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 69: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 5$ is the first whose minimal order is exactly the stated 69.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 218$ exact terms from $n = 5$ on, it returns order 69 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{69} c_i \, a(n-i),$$

c_1	6	c_{19}	−2642668	c_{37}	27118638	c_{55}	−2352583
c_2	−3	c_{20}	1198358	c_{38}	38415606	c_{56}	−1389538
c_3	8	c_{21}	7805672	c_{39}	−16164109	c_{57}	−478597
c_4	−59	c_{22}	7976630	c_{40}	−60462113	c_{58}	123711
c_5	−304	c_{23}	−5151721	c_{41}	−25273807	c_{59}	358798
c_6	−79	c_{24}	−16376066	c_{42}	46588007	c_{60}	313337
c_7	311	c_{25}	−15880620	c_{43}	63186133	c_{61}	222320
c_8	4047	c_{26}	979666	c_{44}	11676515	c_{62}	121631
c_9	8552	c_{27}	37019833	c_{45}	−40150044	c_{63}	54037
c_{10}	−278	c_{28}	39115781	c_{46}	−41924515	c_{64}	16577
c_{11}	−24200	c_{29}	−20902025	c_{47}	−10518210	c_{65}	−2120
c_{12}	−85883	c_{30}	−62870338	c_{48}	13117803	c_{66}	−2638
c_{13}	−88807	c_{31}	−27603319	c_{49}	16167267	c_{67}	−2106
c_{14}	128334	c_{32}	37792570	c_{50}	9503338	c_{68}	−576
c_{15}	473802	c_{33}	53975449	c_{51}	3464349	c_{69}	288
c_{16}	702505	c_{34}	6193128	c_{52}	−232507		
c_{17}	−287234	c_{35}	−40103395	c_{53}	−2239674		
c_{18}	−2478699	c_{36}	−25385710	c_{54}	−2748461		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 74$, and it is the recurrence OEIS A252453 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 5$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{69} - c_1 x^{69-1} - \cdots - c_{69}$. Its constant term is $-c_{69} = -288$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 69 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 69, so $\deg q = 69$, and it is monic. It holds from some index t on. From $\max(t, 5)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 69, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 19 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 69, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 288.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-69 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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